

Distance Calculation in KNN (Complete Notes)

1 Why Distance Calculation is Important in KNN

KNN (K-Nearest Neighbors) classifies a data point based on the **closest neighbors**. So, the **distance metric** determines:

- Which points are “nearest”
- Final classification accuracy

Different **types of data** need different distance measures.

2 Distance Measures for Different Data Types

Types of attributes:

1. **Numeric data**
2. **Categorical (nominal) data**
3. **Heterogeneous data** (numeric + categorical)

3 Distance for Numeric Data

◆ Euclidean Distance

Used when all attributes are numeric.

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Example

Points:

- $x = (2, 4)$
- $y = (5, 8)$

$$d = \sqrt{(2 - 5)^2 + (4 - 8)^2} = \sqrt{9 + 16} = \boxed{5}$$

◆ Normalized Euclidean Distance

Used when numeric attributes have **different scales**.

$$d(x, y) = \frac{|x - y|}{\max - \min}$$

Example

- Age range: 20–60
- Test age = 30, Train age = 50

$$d = \frac{|30 - 50|}{60 - 20} = \frac{20}{40} = \boxed{0.5}$$

Transform raw feature values into z-scores

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

- ▶ x_{ij} is the value for the i^{th} sample and j^{th} feature
- ▶ μ_j is the average of all x_{ij} for feature j
- ▶ σ_j is the standard deviation of all x_{ij} over all input samples

4 Distance for Categorical Data

✗ Problem with Euclidean Distance

For values like Red, Blue, subtraction makes **no sense**.

Information Gain (IG) Formula

Information Gain measures the reduction in entropy after splitting a dataset on an attribute.

Formula:

$$IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} \times Entropy(S_v)$$

Where:

- S = original dataset
- A = attribute used for splitting
- S_v = subset of S where attribute A has value v
- $|S|$ = total number of samples
- $|S_v|$ = number of samples in subset S_v

5 VDM (Value Difference Metric)

- ◆ **Purpose** Used for **categorical attributes** in KNN.
- ◆ **Idea** Two values are similar if they lead to **similar class distributions**.

◆ VDM Formula

$$VDM(v_1, v_2) = \sum_{c \in \text{Classes}} |P(c | v_1) - P(c | v_2)|$$

◆ VDM Example

Dataset

Color	Class
Red	Yes
Red	Yes
Blue	No
Blue	No

Probabilities:

- $P(\text{Yes} | \text{Red}) = 1, P(\text{No} | \text{Red}) = 0$
- $P(\text{Yes} | \text{Blue}) = 0, P(\text{No} | \text{Blue}) = 1$

$$VDM(\text{Red}, \text{Blue}) = |1 - 0| + |0 - 1| = \boxed{2}$$

6 Distance for Heterogeneous Data

◆ What is Heterogeneous Data?

Data with:

- Numeric attributes
- Categorical attributes

7 HVDM (Heterogeneous Value Difference Metric)

◆ Purpose

Used in KNN when **both numeric and categorical attributes exist**.

◆ HVDM Formula

$$HVDM(x, y) = \sqrt{\sum_{\text{numeric}} \left(\frac{|x_a - y_a|}{\text{range}_a} \right)^2 + \sum_{\text{categorical}} VDM(x_a, y_a)^2}$$

8 Complete HVDM Example

Dataset

Instance	Age (Numeric)	Color (Categorical)	Class
X ₁	20	Red	Yes
X ₂	25	Red	Yes
X ₃	30	Blue	No
X ₄	35	Blue	No

Test instance:

$$X_{test} = (28, Red)$$

Step 1: Normalize Numeric Attribute

Range of Age:

$$35 - 20 = 15$$

Step 2: VDM for Color

- $VDM(Red, Red) = 0$
- $VDM(Red, Blue) = 2$

Step 3: Compute HVDM Distances

To X₁ (20, Red)

$$d_{num} = \frac{|28 - 20|}{15} = 0.533$$
$$HVDM = \sqrt{0.533^2 + 0^2} = \boxed{0.533}$$

To X₂ (25, Red)

$$d_{num} = \frac{|28 - 25|}{15} = 0.2$$
$$HVDM = \boxed{0.2}$$

To X₃ (30, Blue)

$$d_{num} = \frac{2}{15} = 0.133$$
$$HVDM = \sqrt{0.133^2 + 2^2} \approx \boxed{2.00}$$

9 KNN Classification (k = 3)

Nearest neighbors:

- X₂ → Yes

- $X_1 \rightarrow \text{Yes}$
- $X_3 \rightarrow \text{No}$

Final Class:

Yes

10 Summary Table (Excellent for Exams)

Data Type	Distance Metric
Numeric	Euclidean / Normalized Euclidean
Categorical	VDM
Mixed (Heterogeneous)	HVDM

Nearest Neighbour (KNN): Computational Complexity (Easy Explanation)

1 Why Nearest Neighbour is Expensive

- To find the nearest neighbour of a new point q , we must calculate the distance between q and every training data point.
- If there are N training examples, we do N distance calculations.
- This makes KNN **slow**, especially for large datasets.

2 How to Make It Faster

- **KD-Trees:**
Store training data in special tree structures to reduce search time.
- **Approximate methods (LSH):**
Find an approximate nearest neighbour instead of the exact one, which is faster.
- **Condensing:**
Remove unnecessary or duplicate data points to reduce computation.

3 Storage Requirements

- KNN must store **all training data points**.
- This needs **a lot of memory** when the dataset is large.
- **Condensing** helps by keeping only important data points.
- **Pre-sorting (like KD-trees)** may increase memory usage because extra structures are stored.

4 High-Dimensional Data Problem

- KNN performs poorly when the number of features (dimensions) is high.

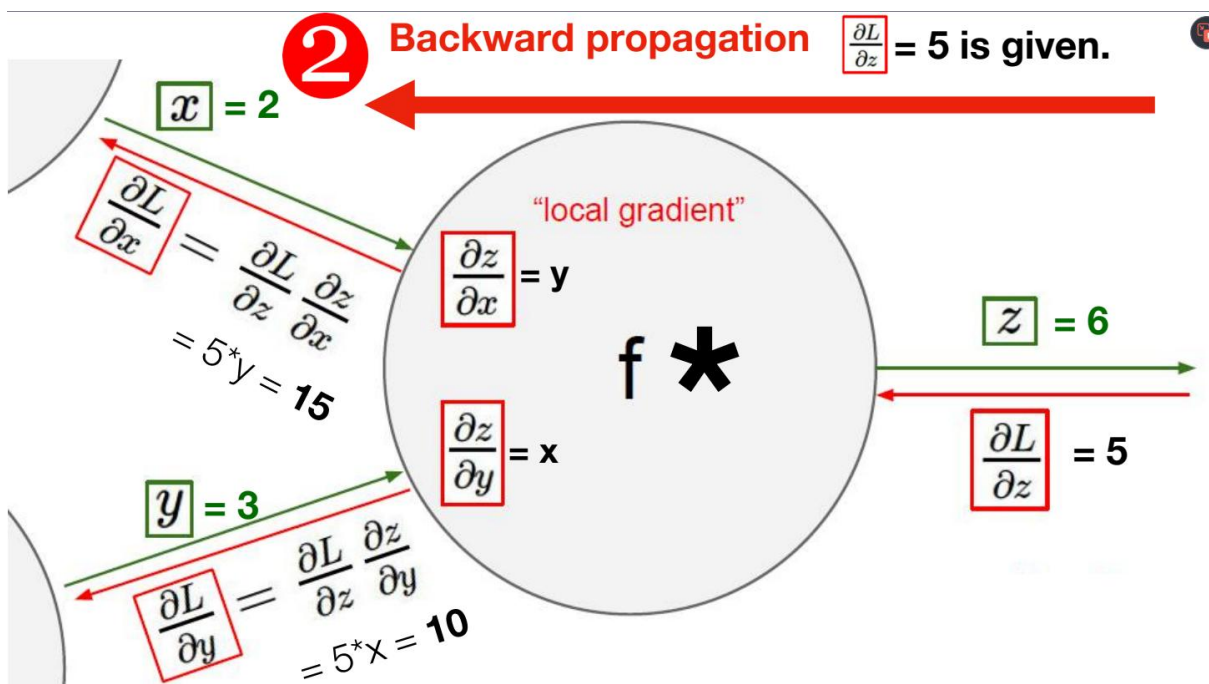
- This problem is called the **Curse of Dimensionality**.
- As dimensions increase:
 - More training data is needed
 - Computation becomes very slow
- In very high dimensions, smart search methods behave like **simple linear search**.

5 One-Line Summary (Exam-Friendly)

- Nearest Neighbour is simple but slow and memory-intensive, especially for large and high-dimensional datasets.

Reduction in Computational Complexity

- Reduce size of training set
- Condensation, editing
- Use geometric data structure for high dimensional search



What this diagram represents

Inside the circle:

$$z = f(x, y) = x \times y$$

Given values:

- $x = 2$
- $y = 3$

So **forward pass**:

$$z = 2 \times 3 = 6$$

That's why the diagram shows **z = 6** 

What is GIVEN in backward pass?

The diagram says:

$$\frac{\partial L}{\partial z} = 5$$

👉 This means:

"If z changes a little, the loss L changes 5 times that amount."

You **do NOT** calculate 5 — it is **given**.

Step 1: Find local gradients (very important)

Because:

$$z = x \times y$$

Local gradient w.r.t. x

$$\frac{\partial z}{\partial x} = y$$

Since $y = 3$:

$$\frac{\partial z}{\partial x} = 3$$

Local gradient w.r.t. y

$$\frac{\partial z}{\partial y} = x$$

Since $x = 2$:

$$\frac{\partial z}{\partial y} = 2$$

These are the **red boxes inside the circle** in the picture.

Step 2: Apply CHAIN RULE (this is the key)

Gradient of loss w.r.t. x

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial x}$$

Substitute values:

$$\frac{\partial L}{\partial x} = 5 \times 3 = 15$$

✅ That's why the diagram shows **15**

Gradient of loss w.r.t. y

$$\frac{\partial L}{\partial y} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial y}$$

Substitute values:

$$\frac{\partial L}{\partial y} = 5 \times 2 = 10$$

✅ That's why the diagram shows **10**

Final results (boxed)

$$\boxed{\frac{\partial L}{\partial x} = 15}$$

$$\boxed{\frac{\partial L}{\partial y} = 10}$$